

Empirical Design, Experiments, And Frontier Questions

From Labor-Market Mechanisms To Applied Research Designs

Special Topic 7: Labor Market Design, Contracting, and Mechanism Design – Week 6

Central question

- How can labor-market design interventions be evaluated credibly?
- The capstone object is not a generic estimator.
- It is mechanism $\rightarrow$ margin $\rightarrow$ counterfactual $\rightarrow$ equilibrium response $\rightarrow$ welfare.
- Students leave with a template for theory-backed applied projects on matching, recruiting, contracts, platforms, and public-sector assignment.

Mechanism-first empirical design

Mechanism	Labor margin	Empirical object
Deferred acceptance	ranking and stability	assigned rank, blocking risk, match quality
Timing rule	offer pressure and congestion	applications, interviews, acceptances
Incentive contract	effort, sorting, multitasking	output, entry, retention, task allocation
Platform ranking	attention and visibility	views, bids, hires, wages
Public priority rule	staffing and legitimacy	hard-to-staff fill, worker welfare

rule $\rightarrow$ behavioral margin $\rightarrow$ labor outcome $\rightarrow$ welfare object

Counterfactuals and credible evaluation

- A design estimate is always relative to another rule or institution.
- Matching: old algorithm, decentralized offers, modified priorities, or capacity changes.
- Recruiting: different timing, application limits, signals, or information.
- Contracts: different monitoring technology, wage rule, bonus formula, or probation design.
- Platforms: ranking, disclosure, recommendation, wage floor, or fee rule.
- Public assignment: different balance of worker preferences, priorities, staffing, and transparency.

Experiments, audits, and pilots

Design	Best use	Main threat
Field experiment Platform A/B test	direct rule or information change ranking, visibility, wage, or message rule	spillovers and anticipation equilibrium reallocation
Audit	screening, discrimination, first-stage response	later match quality unobserved
Institutional pilot	real assignment or staffing reform	bundled implementation changes
Staggered rollout	administrative or legal reform	adoption and trend confounding

- The right question is what the intervention moves and which agents can respond.

Administrative and platform data

- Rich process data reveal margins that final matches hide.
- Applications, rankings, interviews, offers, waitlists, bids, messages, exposure, and acceptances show how the rule works.
- Linked outcomes add wages, productivity, retention, promotions, training, exit, and public-service performance.
- The key advantage is process visibility, not just sample size.
- These data help separate mechanism effects from correlations in realized matches.

Structural matching and equilibrium evaluation

$$y_i(\rho) = d_i(\rho) + b_i(\rho) + e_i(\rho)$$

- $d_i(\rho)$: direct design effect of the rule.
- $b_i(\rho)$: behavioral response by workers, firms, programs, or managers.
- $e_i(\rho)$: equilibrium adjustment through congestion, sorting, wages, or capacity.
- Structural evaluation is useful when welfare depends on unchosen matches or unimplemented mechanisms.
- Reduced-form and experimental evidence are strongest for local mechanisms.
- Best projects make the two speak to each other.

Welfare and portability

- A better metric is not automatically a welfare gain.
- A stable match may lower wages; faster recruiting may worsen fit; public staffing gains may reduce worker welfare.
- Welfare claims must name whose welfare counts and whether surplus rises or rents shift.
- Portability should be argued through the mechanism: congestion, information, incentives, assignment, priority design, or bargaining power.
- External validity fails when outside options, legal constraints, wage setting, or equilibrium response differ in central ways.

Frontier project map

Frontier area	Research question	Data/design
Within-vacancy design	how are applicants screened under bottlenecks?	application-level admin data
Platform wage rules	who bears pricing and transparency changes?	A/B tests, rollouts
Internal assignment	do rules affect promotions and retention?	personnel panels, pilots
Public staffing	can priorities improve service without exit?	assignment and outcome data
AI screening	what margin does automation move?	audit plus process data

- Primary anchor: empirical evaluation of the medical match.
- Challenge anchor: deferred acceptance in Army officer labor markets.
- Reproduce: compare a decentralized early-offer rule with deferred-acceptance-style assignment in synthetic matching data.
- Diagnose: separate preference rank, priority alignment, match quality, shortage staffing, spillovers, and welfare.
- Transfer: write a frontier design memo for platforms, recruiting, contracts, teacher placement, public staffing, internal assignment, or AI screening.

Student project deliverable

Element	What the memo must state
Institution	the labor market and rule being studied
Mechanism	the design object that moves behavior
Labor margin	applications, assignment, wages, effort, retention, or staffing
Empirical leverage	randomization, pilot, rollout, cutoff, or model restriction
Counterfactual	the alternative rule or institution
Equilibrium threat	spillovers, sorting, congestion, or strategic response
Welfare object	workers, firms, public agencies, clients, or communities

Takeaways

- Labor-market design research starts with the mechanism, not the estimator.
- Credible evaluation needs a counterfactual and an observable labor object.
- Experiments, audits, pilots, process data, and structural models see different pieces of the same design problem.
- Welfare requires equilibrium diagnosis and a clear account of who gains or loses.
- The course ends with applied project design: theory-backed, institutionally serious, and empirically disciplined.